Prompt-Calibrated Vision Transformers for Open-Set Recognition with CLIP-Driven Score Fusion

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***Abstract*—Standard image classifiers operate under a closed-set assumption: every test image is forced into one of the training classes. Open-set recognition (OSR) relaxes this assumption by requiring a model to classify samples from known classes while rejecting samples from classes never seen during training. We study OSR with frozen vision–language foundation models. We adapt CLIP to the known classes with a lightweight prompt calibration that learns a residual vector per class text embedding while keeping both CLIP encoders frozen, improving known-class accuracy from 92.38% to 93.54% on a CIFAR-10 open-set split. Our main contribution is a fused open-set score that combines an energy score over the calibrated similarity logits, the cosine-similarity margin between the top two classes, and the distance to the nearest known-class centroid, with fusion weights tuned on a validation set that never touches the test unknowns. The fused score consistently outperforms zero-shot CLIP scoring (0.9493 vs. 0.9205 AUROC on CIFAR-10), while a linear probe with energy scoring on frozen ViT features proves to be an even stronger baseline across all benchmarks. We evaluate on CIFAR-10, CIFAR-100 and TinyImageNet open-set splits using AUROC, FPR@95TPR and OSCR, and provide a failure-case analysis showing that residual errors concentrate on unknowns that share silhouette and background statistics with known classes.**

***Index Terms*—open-set recognition, CLIP, vision transformer, prompt tuning, energy score, out-of-distribution detection**

I. Introduction

Image classifiers deployed in the real world routinely encounter inputs from categories that were not present in their training data. Under the standard closed-set formulation, such inputs are silently mapped to the nearest known class, often with high confidence. Open-set recognition (OSR) [1] addresses this by adding a second requirement: in addition to classifying known-class samples correctly, the model must detect and reject unknown-class samples.

Classical OSR approaches attach a rejection mechanism to a discriminatively trained network, for example by thresholding the maximum softmax probability (MSP) [3] or by recalibrating logits with extreme value theory as in OpenMax [2]. Energy-based scoring [4] improves on MSP by using the log-sum-exp of the logits, which is better aligned with the underlying data density. All of these methods, however, inherit the representation of the closed-set classifier they are attached to.

Vision–language foundation models such as CLIP [7] offer a different starting point. CLIP embeds images and natural-language class descriptions into a shared space, enabling zero-shot classification via cosine similarity, and its representations transfer broadly. Prompt learning methods such as CoOp [8] adapt CLIP to a downstream task by optimizing the text prompts while keeping the encoders frozen, at a small fraction of the cost of fine-tuning.

In this work we ask how far a *frozen* CLIP model can be pushed on OSR with only lightweight prompt calibration and a carefully designed open-set score. Our contributions are:

- A systematic comparison on open-set splits of CIFAR-10, CIFAR-100 and TinyImageNet between a pretrained ViT baseline (with a linear probe on frozen features), zero-shot CLIP, and prompt-calibrated CLIP.
- A lightweight prompt calibration for CLIP that learns one residual vector per class text embedding, trained with temperature-scaled cross-entropy on the known classes only.
- A fused open-set score combining energy, cosine-similarity margin, and centroid distance, with fusion weights selected on a leakage-free validation set using proxy unknowns from a disjoint dataset.
- An evaluation with AUROC, FPR@95TPR and OSCR, together with a qualitative and quantitative failure-case analysis.

II. Related Work

Open-set recognition. Scheirer et al. [1] formalized OSR and introduced the notion of open-space risk. OpenMax [2] replaced the softmax layer with an extreme-value calibration of activation vectors. Hendrycks and Gimpel [3] established MSP as a strong and simple baseline for detecting misclassified and out-of-distribution samples. Liu et al. [4] proposed the energy score $E(x) = -\log \sum_i e^{f_i(x)}$, showing it is theoretically better aligned with the input density than the softmax confidence. Dhamija et al. [5] introduced evaluation practice that

jointly considers classification and rejection, which motivates the OSCR metric used here. Vaze et al. [10] showed that the open-set performance of a network is strongly correlated with its closed-set accuracy, and that a well-trained closed-set classifier is itself a very strong open-set recognizer.

Vision transformers and foundation models. The Vision Transformer [6] applies a pure transformer to image patches and, when pretrained at scale, provides strong transferable features. CLIP [7] trains image and text encoders contrastively on 400M image–text pairs, and performs zero-shot classification by comparing image embeddings against embeddings of class-name prompts.

Prompt learning. CoOp [8] replaces hand-written prompt templates with learnable context vectors optimized on the downstream task while the CLIP encoders stay frozen. Our calibration is a simplified variant that learns a residual directly in the text embedding space, which retains the spirit of prompt learning (frozen backbone, few trainable parameters) with an even smaller parameter count: one d-dimensional vector per known class.

III. Method

A. Problem Setup

Let $\mathcal{C}_k$ be the set of known classes and $\mathcal{C}_u$ a disjoint set of unknown classes. Training uses only samples from $\mathcal{C}_k$. At test time the model receives samples from $\mathcal{C}_k \cup \mathcal{C}_u$ and must (i) classify known samples into $\mathcal{C}_k$ and (ii) reject unknown samples. Rejection is implemented by thresholding an open-set score $s(x)$, where higher values indicate "known".

B. ViT Baseline with Linear Probe

The baseline uses a ViT-B/16 pretrained on ImageNet with all weights frozen. A linear probe $W \in \mathbb{R}^{|\mathcal{C}_k| \times d}$ is trained with cross-entropy on the CLS-token features of the known-class training images. Open-set scores are computed from the probe logits $f(x)$: MSP $= \max_i \operatorname{softmax}(f(x))_i$ and energy $= \log \sum_i e^{f_i(x)}$ (we use the negated convention throughout so that higher means known). The probe is essential: without it, the pretrained 1000-way ImageNet head produces confidence scores that carry no information about the known/unknown split, and the resulting AUROC falls below chance (0.42 in our preliminary experiments).

C. CLIP Zero-Shot OSR

CLIP ViT-B/32 encodes each known class name c with the template "a photo of a $\{c\}$" into a normalized text embedding t_c, and each image into a normalized embedding v. Classification is $\arg\max_c v^\top t_c$. As open-set scores we use the maximum cosine similarity $\max_c v^\top t_c$ and the negative entropy of the softmax over similarities.

D. Prompt Calibration

To adapt CLIP to the known classes without touching its encoders, we learn a residual vector $p_c \in \mathbb{R}^d$ per class and use calibrated text embeddings

$$\tilde{t}_c = \frac{t_c + p_c}{\|t_c + p_c\|}. \tag{1}$$

The p_c are the only trainable parameters ($|\mathcal{C}_k| \times d$, e.g., 5×512 for CIFAR-10). They are trained with cross-entropy on temperature-scaled logits $\tau\, v^\top \tilde{t}_c$, where τ is CLIP's learned logit scale. Temperature scaling is important: raw cosine similarities lie in $[-1, 1]$, so the softmax over them is nearly uniform and gradients nearly vanish. Because the image encoder is frozen, image features are precomputed once, reducing prompt training to seconds.

E. Fused Open-Set Score

Our open-set score fuses three complementary signals computed with the calibrated embeddings:

- **Energy**: $s_E(x) = \log \sum_c e^{\tau v^\top \tilde{t}_c}$, a density-aligned confidence over the calibrated logits.
- **Margin**: $s_M(x) = v^\top \tilde{t}_{(1)} - v^\top \tilde{t}_{(2)}$, the gap between the top-2 similarities; unknowns tend to sit between classes and have small margins.
- **Distance**: $s_D(x) = \max_c v^\top \mu_c$, the cosine similarity to the nearest known-class centroid μ_c computed from training features; a text-free, purely visual consistency check.

Each component is z-normalized using statistics of held-out known validation samples, and the final score is

$$s(x) = \alpha\, \hat{s}_E(x) + \beta\, \hat{s}_M(x) + \gamma\, \hat{s}_D(x), \qquad \alpha+\beta+\gamma = 1. \tag{2}$$

F. Leakage-Free Weight Selection

Tuning (α, β, γ) on the test unknowns would leak information about the evaluation. Instead, we hold out 20% of the known training images as validation knowns and use samples from a *disjoint* dataset as proxy unknowns (CIFAR-100 when evaluating on CIFAR-10 and TinyImageNet; CIFAR-10 when evaluating on CIFAR-100). The weights are selected by grid search over $\{0, 0.25, 0.5, 0.75, 1\}^3$ (normalized to sum to one) to maximize validation AUROC.

IV. Experimental Setup

Datasets and splits. CIFAR-10 with 5 known classes {airplane, automobile, cat, dog, horse} and 5 unknown classes {bird, deer, frog, ship, truck}; CIFAR-100 with the first 20 classes known and the remaining 80 unknown; TinyImageNet with the first 20 classes known and the remaining 180 unknown. Only known-class training images are used for any training.

Models. ViT-B/16 (ImageNet pretrained, timm) for the baseline; CLIP ViT-B/32 (OpenAI weights, OpenCLIP) for all CLIP-based methods. All backbones are frozen in all experiments.

Metrics. AUROC for known-vs-unknown detection; FPR@95TPR, the false positive rate at the threshold accepting 95% of knowns (lower is better); and OSCR [5], the area under the correct classification rate versus false positive rate curve, which jointly rewards correct known-class classification and unknown rejection.

Implementation. Prompt calibration: 10 epochs, Adam, learning rate 10^{-3}, batch size 512, on precomputed features.

Linear probe: 10 epochs, Adam, learning rate 10^{-3}. All experiments run on a single NVIDIA T4 GPU in Google Colab.

V. Results

A. Main Comparison

Table I reports the main comparison. Two findings stand out. First, the fused score consistently improves over zero-shot CLIP scoring on every dataset, with the largest gains where zero-shot scoring degrades (CIFAR-100: 0.8767 vs. 0.7560 entropy AUROC), and yields far better OSCR than zero-shot similarity on TinyImageNet (0.7492 vs. 0.5774) thanks to the higher known-class accuracy of the calibrated model. Second, once the ViT baseline is equipped with a proper linear probe, energy scoring on its logits becomes the strongest method overall, achieving the best AUROC, FPR@95TPR and OSCR on all three benchmarks. This echoes the observation of Vaze et al. [10] that a good closed-set classifier is itself a very strong open-set recognizer, and positions our fusion as the best approach among methods that require no supervised head training.

B. Ablation of the Fused Score

Table II isolates the contribution of each score component on CIFAR-10. The energy score is the strongest individual component, while margin and distance are weaker alone but complementary: the tuned fusion matches the best pairwise combination (0.9447 vs. 0.9455) without ever accessing the test unknowns. The ablation uses an independent training run of the prompt calibration; the small difference from Table I (0.9447 vs. 0.9493) reflects run-to-run variance in prompt initialization.

C. Effect of Prompt Calibration

Prompt calibration improves known-class accuracy from 92.38% to 93.54% on CIFAR-10 while training only 5×512 parameters. Table III isolates its effect on each score component. The gain is concentrated almost entirely in the energy score (0.9026 $\rightarrow$ 0.9443 AUROC): calibration sharpens the similarity logits, which the temperature-scaled log-sum-exp amplifies. The margin is nearly unchanged and the centroid distance is exactly unchanged, as expected, since it depends only on image features and not on the text embeddings that calibration modifies.

VI. Analysis

A. Feature-Space Structure

Figure 1 shows a t-SNE projection of the CLIP image features. Known classes form compact clusters while unknown samples scatter between and around them, which explains why both the margin and the centroid-distance components carry signal.

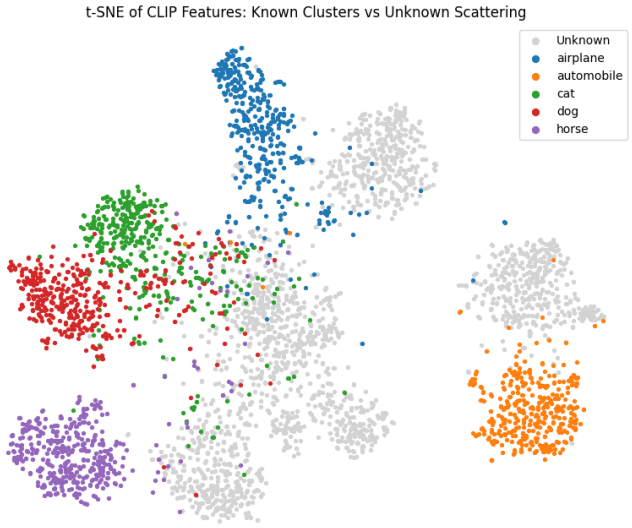


Fig. 1. t-SNE of CLIP features: known classes cluster; unknowns (gray) scatter between clusters.

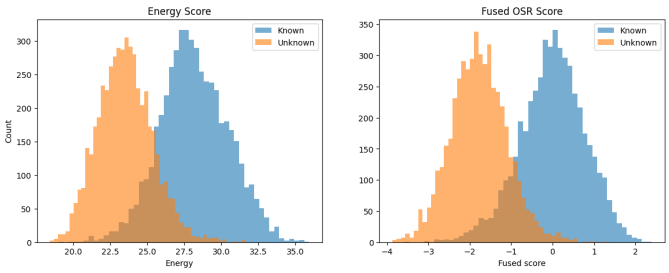


Fig. 2. Known vs. unknown distributions of the energy score (left) and the fused score (right) on CIFAR-10.

B. Score Distributions

Figure 2 shows the distributions of the energy and fused scores for known versus unknown test samples. The fused score yields two well-separated modes with a limited overlap region that contains the genuinely ambiguous samples.

C. Failure Cases

With the rejection threshold set to accept 95% of known samples, the residual errors are highly structured rather than random. Frog is by far the leakiest unknown class (48.2% accepted, almost always predicted as cat), followed by truck (28.1%, predicted as automobile); ship is rejected most reliably (14.2% accepted), with bird (16.2%) and deer (17.2%) in between. Deer confusions split between cat and horse. The pattern is one of semantically and visually plausible confusions between animal classes and between vehicle classes. Figure 3 shows the individual unknown images accepted with the highest confidence; these are dominated by ships and birds photographed against blue sky and predicted as airplane — low-frequency but high-confidence silhouette confusions, complementary to the high-frequency frog–cat leak in the aggregate statistics.

TABLE I
OPEN-SET RECOGNITION RESULTS. HIGHER AUROC/OSCR AND LOWER FPR@95TPR ARE BETTER. BEST VALUE PER DATASET AND METRIC IN BOLD.

Dataset	Method	AUROC	FPR@95	OSCR
CIFAR-10	ViT + MSP	0.9172	0.5482	0.8975
	ViT + Energy	**0.9575**	**0.2600**	**0.9253**
	CLIP + Similarity	0.9153	0.3616	0.8544
	CLIP + Entropy	0.9205	0.4302	0.8624
	Prompt CLIP + Fused (ours)	0.9493	0.2706	0.9229
CIFAR-100	ViT + MSP	0.8852	0.5426	0.8485
	ViT + Energy	**0.9075**	**0.4632**	**0.8592**
	CLIP + Similarity	0.8232	0.7130	0.7185
	CLIP + Entropy	0.7560	0.8094	0.6633
	Prompt CLIP + Fused (ours)	0.8767	0.5871	0.8222
TinyImageNet	ViT + MSP	0.9527	0.3188	0.8592
	ViT + Energy	**0.9687**	**0.1968**	**0.8632**
	CLIP + Similarity	0.9357	0.2760	0.5774
	CLIP + Entropy	0.8038	0.7630	0.5226
	Prompt CLIP + Fused (ours)	0.9339	0.2810	0.7492

TABLE II
ABLATION OF THE FUSED SCORE ON CIFAR-10 (AUROC). TUNED WEIGHTS: $(\alpha, \beta, \gamma) = (0.50, 0.25, 0.25)$.

Configuration	AUROC
Energy only	0.9253
Margin only	0.8626
Distance only	0.8697
Energy + Margin	0.9103
Energy + Distance	**0.9455**
Margin + Distance	0.9356
Fused (tuned weights)	0.9447

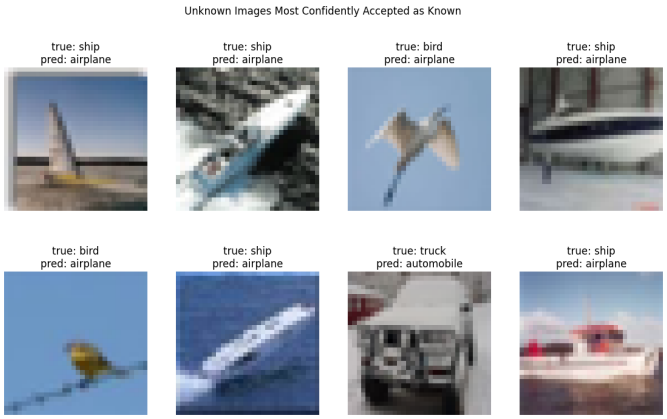


Fig. 3. Unknown test images most confidently accepted as known, with true and predicted labels.

VII. LIMITATIONS

Our study is limited to frozen backbones and small-resolution benchmarks; results may differ under distribution shift or with larger CLIP variants. The proxy-unknown validation strategy assumes access to some disjoint auxiliary data for weight tuning. The fusion weights are selected by coarse grid search; learned fusion may perform better. Finally, unknowns that are visually near-duplicates of known classes (e.g., ship vs. airplane silhouettes at 32×32) remain a fundamental challenge for purely visual OSR scores.

VIII. CONCLUSION AND FUTURE WORK

We showed that a frozen CLIP model, adapted with a per-class residual prompt calibration and equipped with a fused open-set score combining energy, similarity margin and centroid distance, provides a lightweight open-set recognizer that consistently outperforms zero-shot CLIP scoring, while a linear probe with energy scoring on frozen ViT features proves to be an even stronger baseline. Future work includes test-time adaptation of the prompts, open-vocabulary expansion where candidate unknown class names are added dynamically, and synthetic unknown generation in feature space to improve threshold calibration.

TABLE III
AUROC OF EACH SCORE COMPONENT WITH ZERO-SHOT VS. PROMPT-CALIBRATED TEXT EMBEDDINGS (CIFAR-10).

Score	Zero-shot	Prompt-calibrated
Energy	0.9026	**0.9443**
Margin	0.8647	0.8651
Distance	0.8697	0.8697

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